

Moving the Scale for Obesity Therapeutics – From Bench to Bedside: Functional Cell-based Assays for GLP-1, GIP, & Beyond



Ruchi Singhal, Gianni Guizzunti, Shruti Kumar, Venkatesh Chari, Gaurav Agrawal and Jane E. Lamerdin
Eurofins DiscoverX, Fremont, CA 94538

Abstract

To address the escalating global burden of metabolic diseases, the development of effective therapeutics is more critical than ever. GPCR-based peptide therapeutics, particularly those targeting receptors such as GLP-1, GIP, glucagon (GCG), and others, have shown significant potential in combating obesity and diabetes. The innovation in engineered peptides, biosimilars, and small molecules hinges on capturing precise mechanisms of action and robust pharmacology.

Eurofins DiscoverX® provides a comprehensive suite of fit-for-purpose assays, utilizing industry-validated Enzyme Fragment Complementation (EFC) technology to evaluate therapeutics targeting these crucial receptors. These assays evaluate various molecular mechanisms of action (MOAs), including cAMP accumulation, β -arrestin recruitment, and receptor internalization. In this poster, we present data on ready-to-use bioassays, qualified with therapeutics like tirzepatide and retatrutide, which accurately reflect the MOAs of dual and triple receptor peptide analogs. These bioassays are not only fit-for-purpose for quality control (QC) lot-release testing, but also accelerate time-to-market by 6 to 9 months, supporting global screening, characterization, and potency testing programs.

Accelerating Obesity Therapeutic Drug Discovery

Therapeutic Modalities	MOA-reflective Assays	Key Targets	Product Formats
- Small Molecules - Innovator Biologics - Biosimilars, Biobetters & Generics	- cAMP Accumulation β-Arrestin Recruitment - GPCR Internalization	GLP-1, GIP, Glucagon (GCG), GPR75, GPR10, PYY , GPR40, & Amylins	- Thaw & Use Bioassays - Stable Cell Line - Working Cell Banks
Industry Accepted	Supporting Global Programs	Qualified with	Commercial Drug Release
50+ Publications, Patents, & Regulatory Filings	25+ Biologics & Biosimilar Programs	- Semaglutide - Tirzepatide - Retatrutide	✓ ICH-based Qualified Bioassays ✓ Custom Assay Qualification ✓ Global Method Transfer Support ✓ Long-Term Support Commitment

ICH = International Council for Harmonization

Functional, Cell-based Assays for Glucose Metabolism Targets for Multiple MOAs

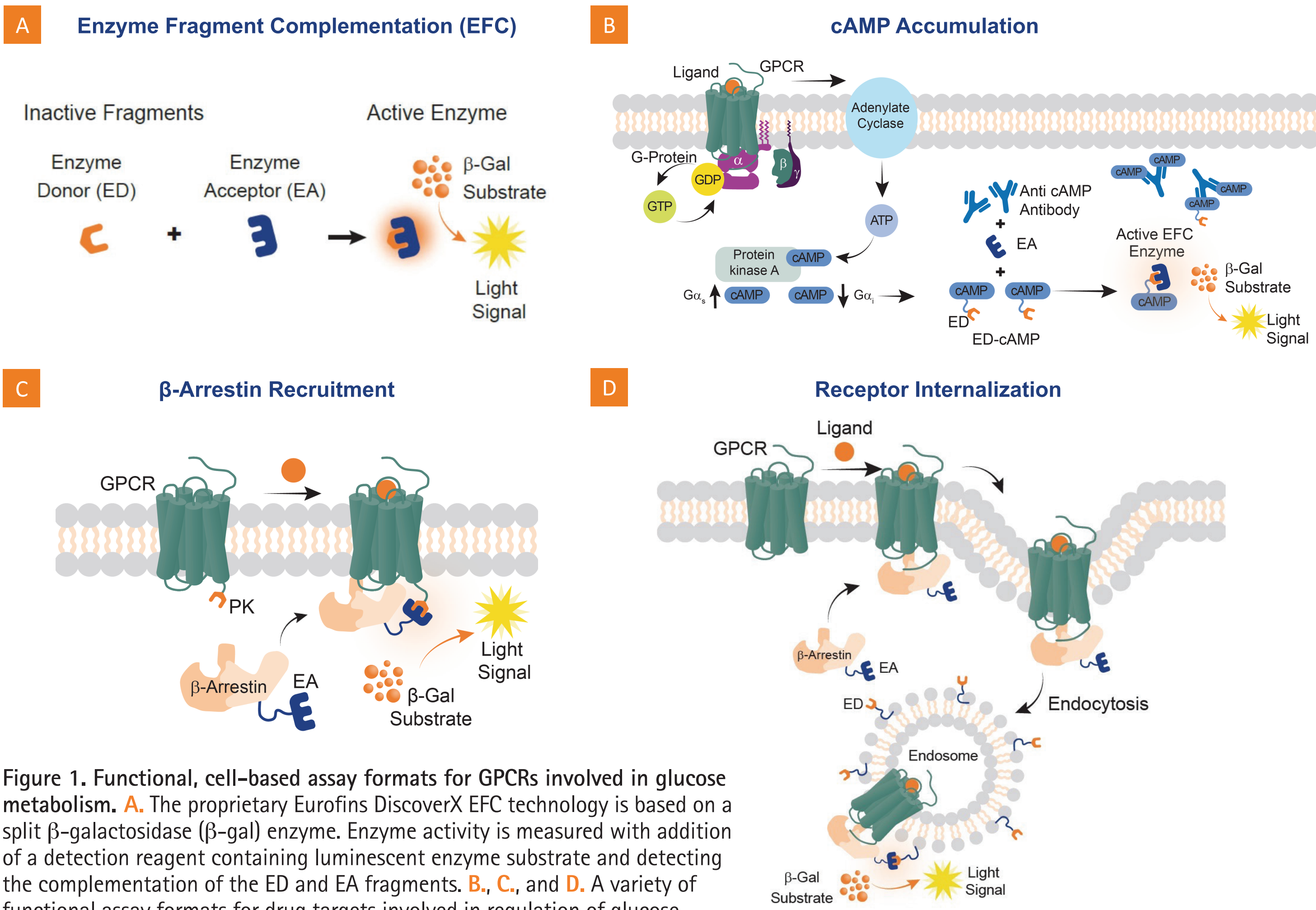


Figure 1. Functional, cell-based assay formats for GPCRs involved in glucose metabolism. **A.** The proprietary Eurofins DiscoverX EFC technology is based on a split β -galactosidase (β -gal) enzyme. Enzyme activity is measured with addition of a detection reagent containing luminescent enzyme substrate and detecting the complementation of the ED and EA fragments. **B., C., and D.** A variety of functional assay formats for drug targets involved in regulation of glucose metabolism for evaluating multiple drug MOAs of glucagon receptor family targets (e.g. GLP-1R, GLP-2R, GIPR, and GCGR). Assays are available as cell lines or bioassays to evaluate activation of second messenger pathways (e.g. cAMP), β -arrestin recruitment, and β -arrestin activated receptor internalization.

Qualification Data Summary for Tirzepatide in GLP-1 RA and GIP RA Bioassays: Demonstrating Suitability for QC-lot Release

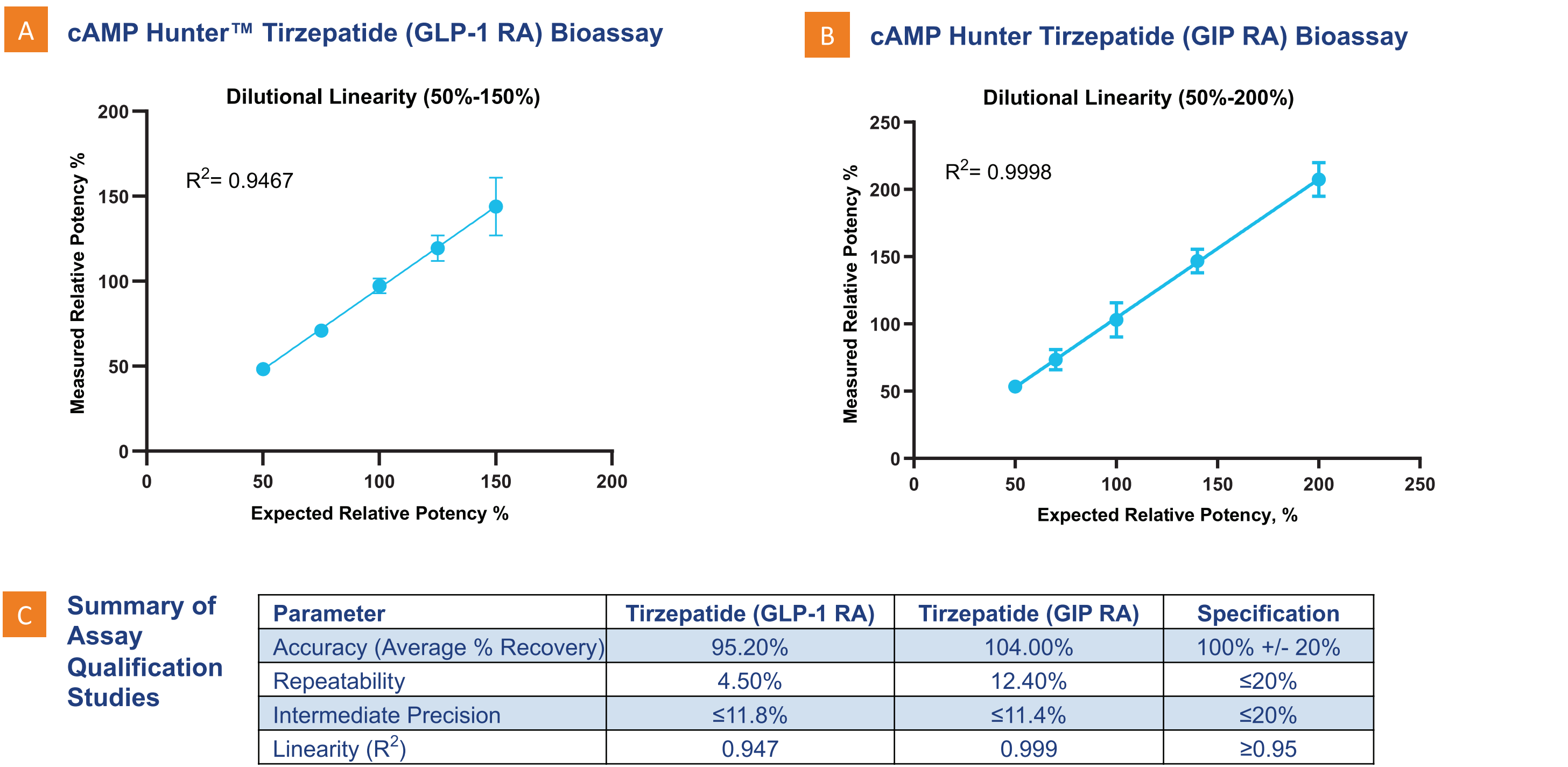


Figure 2. Qualification data summary for the tirzepatide (GLP-1 RA and GIP RA) bioassay. To demonstrate the dilutional linearity of the assay, observed values for each RP were plotted against each "expected" RP, and the slope and R² value of the regression line were calculated. **A.** Dilutional linearity for GLP-1 RA bioassay, R²=0.95. **B.** Dilutional linearity for GIP RA bioassay, R²=0.99. **C.** Results summarized in table, demonstrating good accuracy, repeatability, intermediate precision, and dilutional linearity for the tirzepatide bioassay kits. Five nominal concentrations (NCs) of tirzepatide over a range of 50% to 200% were evaluated (n ≥ 6 for each NC). Repeatability was evaluated using the 100% NC run by a single analyst in one day. Intermediate precision incorporated 2 analysts, multiple days, and evaluation of 2 different lots of bioassay cells. For this study, no plates failed pre-established system and sample suitability criteria for the assay. RP = relative potency. RA = receptor agonist.

Summary

- Eurofins DiscoverX offers a comprehensive portfolio of functional, MOA-reflective cell-based assays that are robust, sensitive, and fit-for-purpose for unimolecular peptides and small molecules.
- Establish accurate pharmacology for your drug using key GPCR readouts including cAMP accumulation, β -arrestin recruitment, and receptor internalization for all targets (GLP-1R, GIPR, and GCGR).
- Accelerate your drug release program into QC lot release with bioassays qualified with approved therapeutics like tirzepatide and retatrutide.
- Consider IND-enabling characterization and assay qualification studies with clinical molecules using Eurofins DiscoverX's custom assay development program (DiscoverXCustomDevelopment@discovery.eurofinsus.com).

Learn more about Eurofins DiscoverX's obesity/diabetes product solutions at discoverx.com/therapeutic-area/obesity-diabetes/

Relative Potency Data for Tirzepatide Using the GLP-1 RA and GIP RA Bioassays

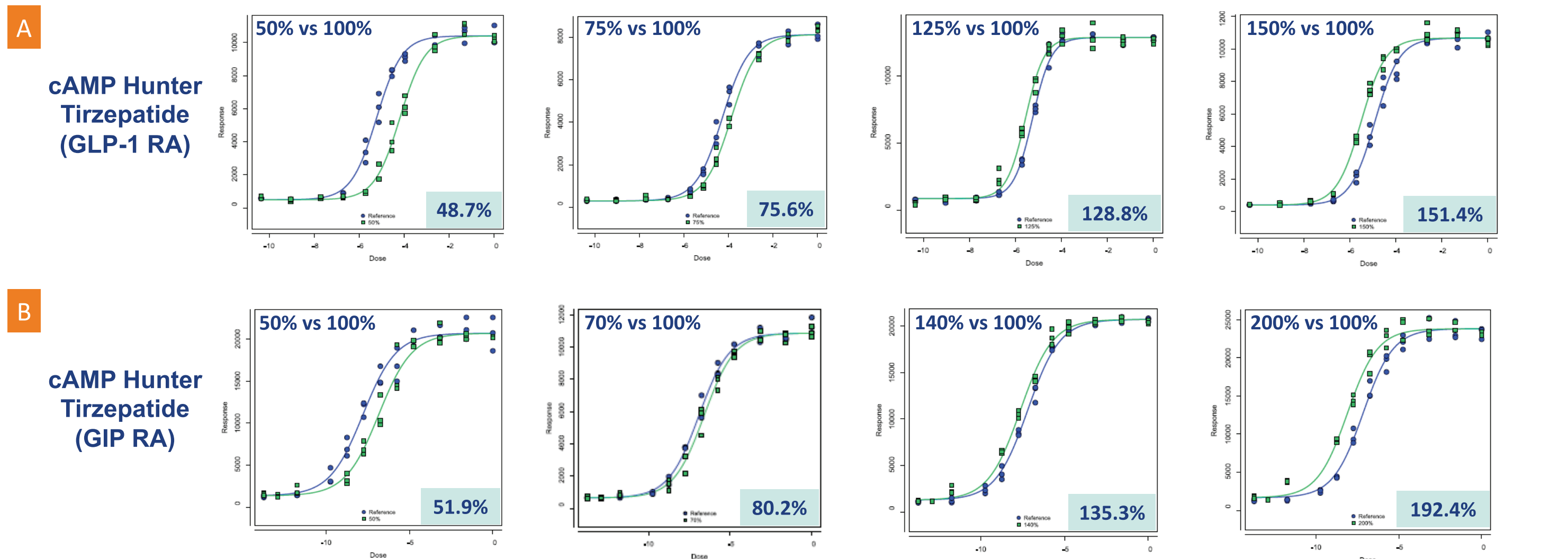
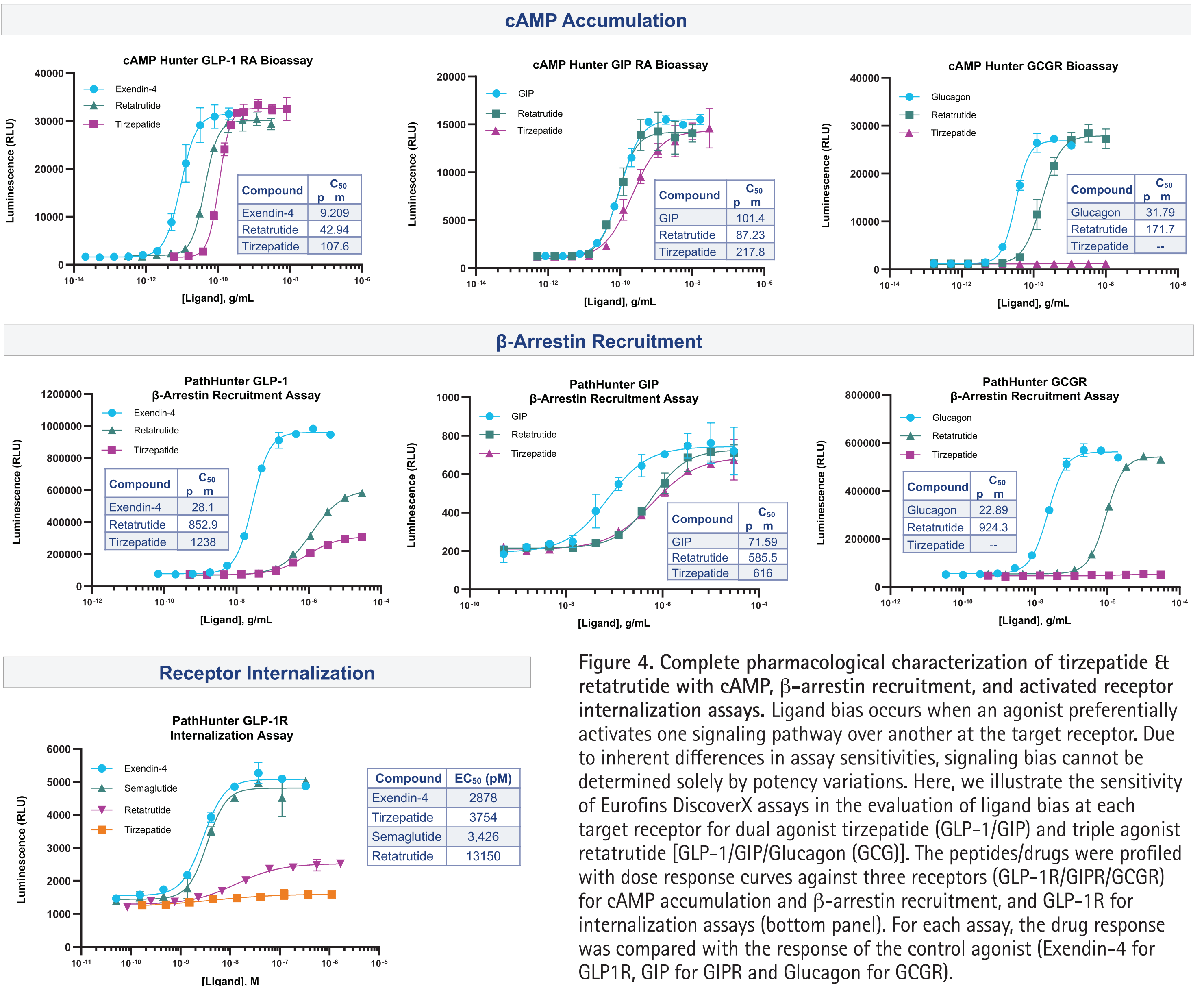


Figure 3. Relative potency data for cAMP Hunter GLP-1 RA and GIP RA bioassays using tirzepatide. **A.** Relative potency data for 100% nominal concentration (NC) of tirzepatide in the cAMP Hunter Tirzepatide GLP-1 RA Bioassay. Samples were tested at varying nominal relative potencies (RP, ranging from 50% to 150%) relative to a reference sample of 100% potency. **B.** Relative potency data for 100% NC of tirzepatide in the cAMP Hunter Tirzepatide GIP RA Bioassay. Samples were tested at 50% to 200%, relative to a reference sample of 100% potency. Each nominal RP was assessed in at least three experiments by two analysts (totaling six assessments per potency level) over multiple days. Data, analyzed using PLA 3.0 software (Stegmann Systems), based on a 95% confidence interval around the 4-parameter logistic fit model. The blue curve represents the reference standard, and the green curve represents the 100% test sample. Relative potency (%) is shown for each graph.

GPCR Biased Signaling Evaluation of Dual and Triple Hormone Receptor Agonists: Tirzepatide and Retatrutide



Inducible GLP-1R cAMP Cell Lines: A Powerful Tool to Study Effects of Receptor Density in Signaling Pathways

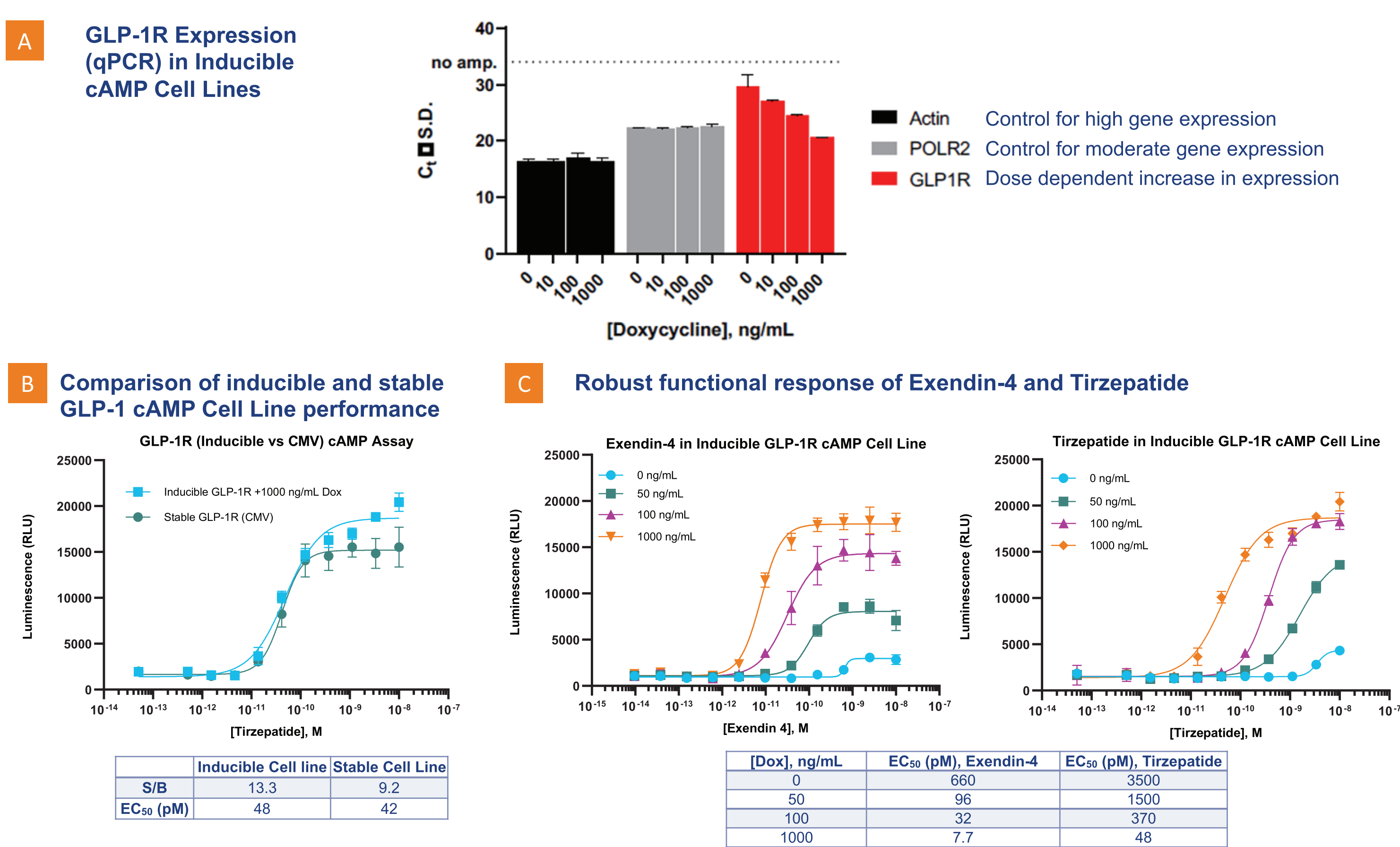


Figure 5. Inducible GLP-1R cAMP cell lines designed to express the GLP-1 receptor in a controllable manner upon treatment with doxycycline. **A.** Effect of different concentrations of doxycycline (0, 10, 100, and 1000 ng/mL) on the expression levels of three genes: actin, POLR2, and GLP-1R. **B.** Comparison of Tirzepatide response in both cell lines, demonstrating comparable signal-to-background (S/B) ratios and EC50s. **C.** Dose response curves for exendin-4 and tirzepatide in inducible cells treated with varying concentrations of doxycycline [Dox]: 0, 50, 100, and 1000 ng/mL